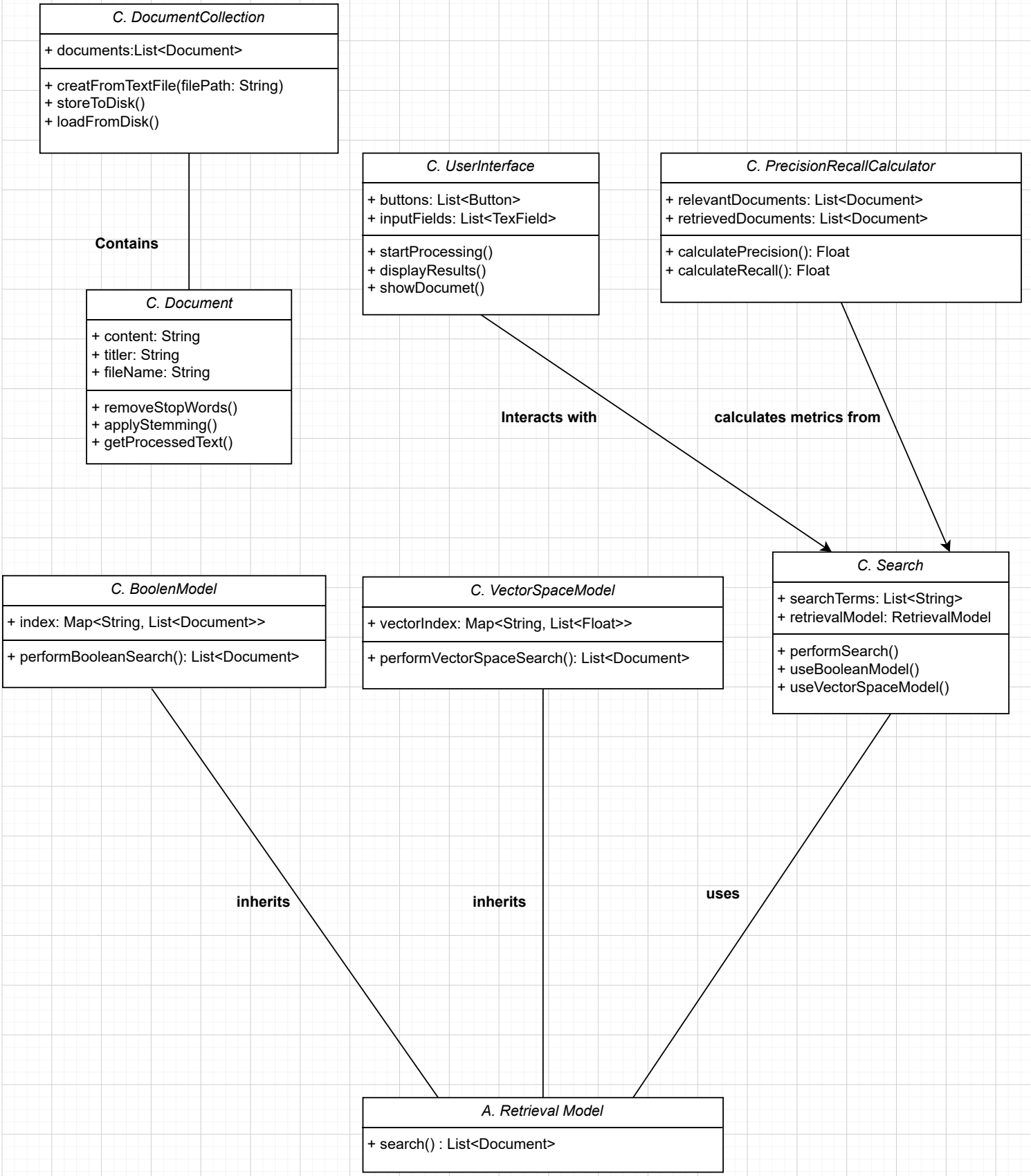
C. DocumentCollection
+ documents:List<Document>
+ creatFromTextFile(filePath: String)
+ storeToDisk()
+ loadFromDisk()
Contains
C. Document
+ content: String
+ titler: String
+ fileName: String
+ removeStopWords()
+ applyStemming()
+ getProcessedText()
C. UserInterface
+ buttons: List<Button>
+ inputFields: List<TexField>
+ startProcessing()
+ displayResults()
+ showDocumet()
C. PrecisionRecallCalculator
+ relevantDocuments: List<Document>
+ retrievedDocuments: List<Document>
+ calculatePrecision(): Float
+ calculateRecall(): Float
Interacts with
calculates metrics from
C. Search
+ searchTerms: List<String>
+ retrievalModel: RetrievalModel
+ performSearch()
+ useBooleanModel()
+ useVectorSpaceModel()
C. BoolenModel
+ index: Map<String, List<Document>>
+ performBooleanSearch(): List<Document>
C. VectorSpaceModel
+ vectorIndex: Map<String, List<Float>>
+ performVectorSpaceSearch(): List<Document>
inherits
inherits
uses
A. Retrieval Model
+ search() : List<Document>

The UML diagram represents an Information Retrieval System with following key components:

Main Components (Classes):

1. DocumentCollection:
   - Attributes: documents: List<Document>
   - Methods:
     - createFromTextFile(filePath: String)
     - storeToDisk()
     - loadFromDisk()
   - Role: Manages a collection of documents. Responsible for loading and storing the document dataset.

2. Document:
   - Attributes: content, title, filename
   - Methods:
     - removeStopWords()
     - applyStemming()
     - getProcessedText()
   - Role: Represents a single document and supports text preprocessing operations.

3. BooleanModel (inherits from RetrievalModel):
   - Attributes: index: Map<String, List<Document>>
   - Method: performBooleanSearch(): List<Document>
   - Role: Implements the Boolean retrieval strategy.

4. VectorSpaceModel (inherits from RetrievalModel):
   - Attributes: vectorIndex: Map<String, List<Float>>
   - Method: performVectorSpaceSearch(): List<Document>
   - Role: Implements the vector space retrieval strategy.

5. Search:
   - Attributes: searchTerms: List<String>, retrievalModel: RetrievalModel
   - Methods:
     - performSearch()
     - useBooleanModel()
     - useVectorSpaceModel()
   - Role: Acts as the controller class for performing searches using different models.

6. PrecisionRecallCalculator:
   - Attributes: relevantDocuments, retrievedDocuments
   - Methods:
     - calculatePrecision(): Float
     - calculateRecall(): Float
   - Role: Computes performance metrics to evaluate search quality.

7. UserInterface:
   - Attributes: buttons: List<Button>, inputFields: List<TextField>
   - Methods:
     - startProcessing()
     - displayResults()
     - showDocument()
   - Role: Allows users to interact with the system.

8. RetrievalModel (Abstract Class):
   - Method: search(): List<Document>
   - Role: Defines the interface for any retrieval model. Used as a base class.

Relationships & Interactions:

- Inheritance:
  - BooleanModel and VectorSpaceModel inherit from the abstract class RetrievalModel.

- Association:
  - Search uses a RetrievalModel to perform search.
  - PrecisionRecallCalculator is associated with Document to compute metrics.
  - UserInterface interacts with both Search and Document to provide feedback and control.

- Encapsulation:
  - Each class defines specific attributes and methods related to its role in the system.